

Snap-HyRE: When Generated Legal Queries Help Legal RAG

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Abstract

Task effectiveness in the legal domain necessitates mapping real scenarios to legal doctrines along with evidential grounding. Furthering this idea, case descriptions may differ from their underlying legal principles being tested: discussing banana peels and brake failure while the legal doctrine required may be about proximate cause or negligences. We introduce Snap-HyRE, a two-call method inspired by HyDE (Hypothetical Document Embeddings) that mirrors expert legal reasoning. The first call writes a private draft answer (an intuition about the controlling doctrine), followed by a legal search passage. This passage is used for retrieval, and the draft is withheld from the second call so the final answer is judged against the corpus, not the guess. On BarExamQA, Snap-HyRE improves raw-question RAG by up to 5.1 percentage points across three model sizes, raises top-five evidence retrieval from 1.4% to 9.5–12.1%, exceeds a single labeled-gold-passage control on Gemma 4 26B, and approximately matches it on Llama 3.3 70B. On HousingQA, where the question already shares vocabulary with the target statute, raw-question RAG remains the stronger answer baseline: Snap-HyRE trails raw by 1.1–3.3 points across the three sizes, while the Gemma 26B row still improves top-five evidence exposure above raw search. A single in-prompt statute example improves Gemma 26B $N = 500$ retrieval probes, but does not make generated queries universally better than raw search. The lift tracks the question–corpus vocabulary gap.

1. Introduction

Legal RAG should work well for legal question answering: many answers turn on statutes, regulations, court holdings, or compact doctrinal rules, and retrieval can put that text in front of a language model (Lewis et al., 2020). The hard part is often the first query. A bar-exam fact pattern may mention a driver, crutches, and a banana peel, while the support passage states a well-established rule about proximate cause. A housing question may name a state, while a national corpus that mixes statutes from every state can return a plausible-sounding rule from the wrong jurisdiction. Final answer accuracy alone hides these failures.

A common fix is to let the model first write the kind of passage it expects to find and search with that text instead of the raw question. HyDe asks the model to generate a synthetic passage that could be used to answer the query, and embeds that text for retrieval of real passages (Gao et al., 2023). Query expansion and rewriting use related generated search text (Wang et al., 2023). The idea is attractive in law because a model can often name the legal category behind a fact pattern before retrieval sees the corpus. These same methods can also fail in this reframe/rewrite step by abstracting away anchors, choosing the wrong legal frame, or retrieving plausible but irrelevant authority.

We introduce Snap-HyRE, a fixed two-call generated-query method that tries to simulate expert legal reasoning. A practitioner facing a fact pattern forms a quick intuition about what is at issue, looks up the controlling authority by that intuition, and then judges the evidence on its own merits rather than anchoring on the first guess. Snap-HyRE encodes the same loop. The first model call writes two objects: (1) a private draft answer (the “snap” intuition) and afterwards (2) a separate legal search passage in the style of the authority that would resolve the question. Only

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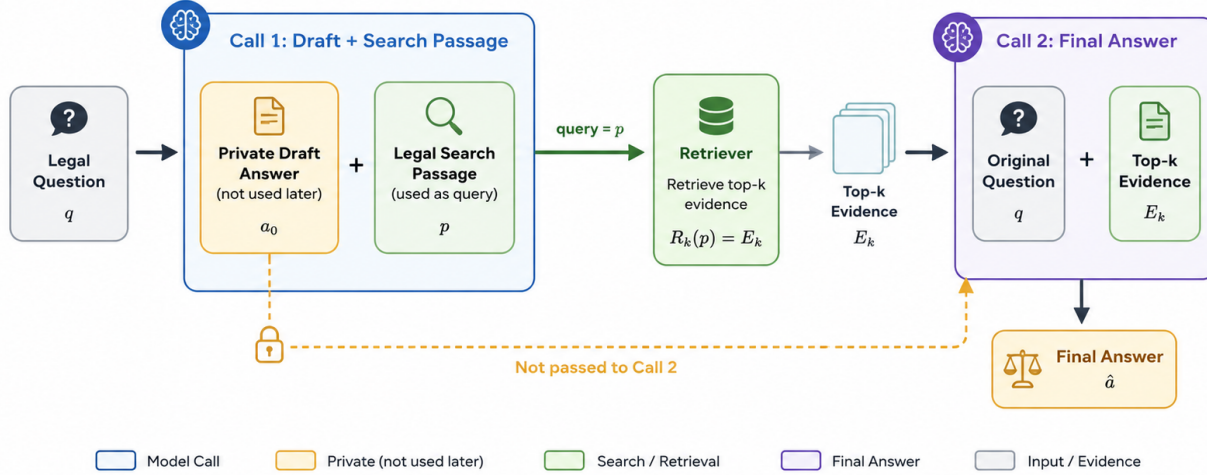


Figure 1. Snap-HyRE: the first model call writes a private draft answer and a separate legal search passage; only the search passage is used to retrieve evidence. The second call sees only the original question and the top- k retrieved passages, and produces the final answer. The draft is never passed to the second call.

the search passage is embedded for retrieval and the draft answer is discarded. The second call answers the original question from the retrieved legal text, judging the evidence without anchoring on its own first guess. The name reflects this structure: “snap” is the quick private draft answer, and HyRE abbreviates Hypothetical Reasoning Embeddings, the legal passage embedded for retrieval.

We report a two-benchmark study chosen for opposite ends of the question-corpus vocabulary gap. BarExamQA is the weak-query regime: fact-pattern questions describe accidents, transactions, and procedures in narrative prose, while the support corpus uses doctrinal vocabulary (“proximate cause,” “hearsay exception,” “adverse possession”). The raw question and the relevant rule passage have low lexical similarity. HousingQA is the strong-query regime: statutory-entailment questions ask directly about landlord and tenant terms (“Can a landlord evict a tenant for committing waste?”), so the question vocabulary already overlaps with the housing statutes that contain the answer.

Snap-HyRE helps in proportion to that gap. On BarExamQA it lifts the share of questions whose labeled gold passage appears in the top five retrieved

results (Hit@5) from 1.4% to 9.5–12.1% across the three model sizes, and improves final answer accuracy at every size. On HousingQA, where raw queries are already corpus-shaped, the boundary is different: raw-question RAG remains strongest and Snap-HyRE trades a 1.1–3.3pp answer margin for higher Gemma 26B evidence exposure than raw search. Section 5 reports the full numbers.

We define Snap-HyRE as a fixed two-call legal RAG method, compare it against raw-question RAG, HyDE, and a Gold Evidence Control that supplies the labeled gold passage directly to the LLM on BarExamQA and HousingQA, and report when the method helps by tracking both answer accuracy and whether gold evidence appears in the top five retrieved passages. Structured query rewriting and additional diagnostics are in the appendix.

2. Background and Related Work

Generated search text. RAG combines a generator with an evidence store: a searchable corpus of text that the system retrieves from before answering (Lewis et al., 2020). For legal tasks the goal is to surface the controlling authority—which are the legal statutes/precedents needed to resolve a dispute—for a given

question, but the question and the corpus often use different language. HyDE addresses this by generating a hypothetical document and embedding that text for retrieval (Gao et al., 2023). Query2doc and related expansion methods similarly use a language model to turn sparse query text into richer retrieval handles (Wang et al., 2023). For legal tasks, the useful generated text is usually not a longer version of the question. It is a passage in the style of the authority the retriever should find.

Legal RAG evaluation. Legal evaluation benchmarks and corpora have standardized many legal-understanding tasks (Guha et al., 2023; Chalkidis et al., 2022; Henderson et al., 2022), and recent work has emphasized the risk of unsupported legal assistant answers (Magesh et al., 2024). Legal RAG benchmarks such as LegalBench-RAG and Legal RAG Bench argue for evaluating retrieval and answering together (Pipitone & Alami, 2024; Butler & Butler, 2026). We follow that joint evaluation but split the failure analysis into two questions: did the retriever miss the evidence, or did the answer model misuse it?

3. Snap-HyRE

3.1. Design Rationale

Snap-HyRE mirrors a small piece of how an expert reads a legal scenario. A practitioner facing a fact pattern forms a quick intuition about what is at issue (e.g. proximate cause, a hearsay exception, or a state-specific notice rule), looks up the controlling authority by that intuition, and then judges the evidence without allowing intuition to bias the judgment. Snap-HyRE encodes the same loop. The first call asks the model to form an initial draft answer from its priors and to write a separate legal search passage for retrieval. The second call reasons only over the question and the retrieved evidence; it never sees the draft answer, so the final answer is judged against what the corpus actually says. Figure 1 shows the resulting two-call pipeline.

3.2. Two Calls, One Retrieval Query

Snap-HyRE separates query formation from final answering. The same model G_θ is used for both calls; only the prompt template differs (the superscript). Given a formatted legal question q , the first call produces a private draft answer a_0 and a legal search passage p :

$$(a_0, p) = G_\theta^{\text{snap}}(q). \quad (1)$$

Algorithm 1 Snap-HyRE inference

Require: question q , model G_θ , retriever R_k

- 1: $(a_0, p) \leftarrow G_\theta^{\text{snap}}(q)$
 - 2: Validate that a_0 and p parse; discard a_0
 - 3: $E_k \leftarrow R_k(p)$
 - 4: $\hat{a} \leftarrow G_\theta^{\text{answer}}(q, E_k)$
 - 5: return $\hat{a}$
-

The retriever R_k embeds only p and returns the top- k retrieved passages $E_k = R_k(p)$. The final answer is then

$$\hat{a} = G_\theta^{\text{answer}}(q, E_k), \quad (2)$$

with a_0 withheld from the final context. The first call can use the model’s prior knowledge of law to choose vocabulary, and the second call must answer from the retrieved corpus text.

Each Snap-HyRE response follows three steps: validate that the first call produced both a formatted draft answer and a clean search passage; retrieve and rerank using only the search passage; answer the original question from the retrieved passages without exposing the draft answer. The retrieval passage is constrained to reflect the corpus it is searching over. Typically describing the controlling rule, statutory condition, legal relationship, or doctrinal test without answer labels or an Answer: header.

HyDE writes a legal passage directly from the question, while our method Snap-HyRE first forms a tentative legal answer and uses that private judgment only to decide what kind of passage to search with. For a torts fact pattern about a broken leg, crutches, and a later fall, this turns an accident-heavy query into a passage about proximate cause and foreseeable subsequent injury (Figure 2).

3.3. Grounding Guardrails

The two-call structure gives the method two guardrails that follow from the formulation above. First, the draft a_0 is private: it can shape what to look for, but it is not handed in as evidence, so the second call avoids the anchoring bias of treating its own first guess as a fact. Second, the search passage p is written like legal authority rather than chain-of-thought reasoning: it names the rule or statutory condition that would resolve the question, not the steps the model used to get there. The intuition step is allowed to use model priors and the evidence step is forced back to the corpus.

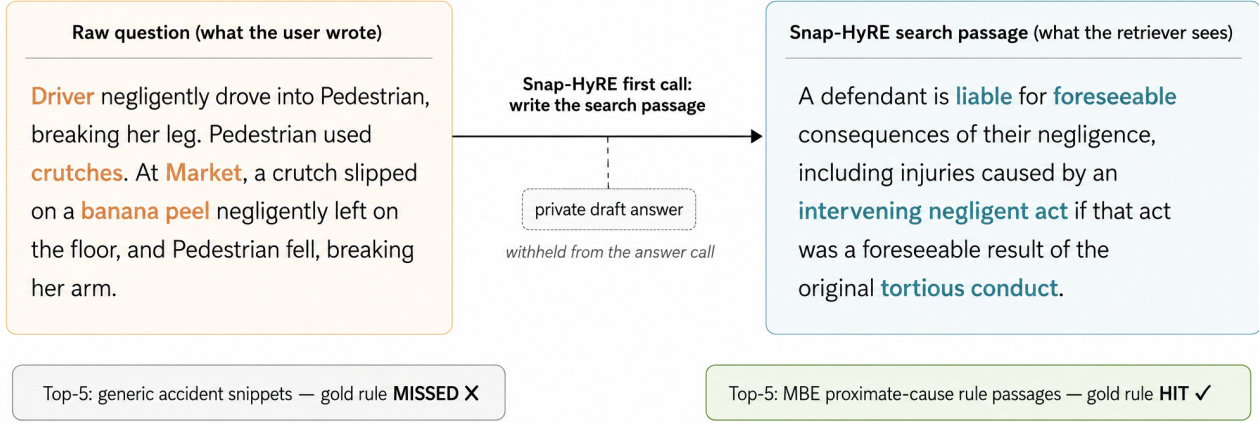


Figure 2. What the two methods send to the retriever for one BarExamQA item (mbe_1175). Raw-question RAG embeds the narrative fact pattern; Snap-HyRE embeds a short doctrinal passage about foreseeable intervening injury, which is the language of the rule the retriever actually needs to surface.

4. Evaluation

4.1. Two Legal Retrieval Regimes

BarExamQA and HousingQA sit at opposite ends of the question-corpus vocabulary gap. BarExamQA questions are real life scenarios that must be interpreted in the context of the law, and the support corpus uses doctrinal vocabulary (Zheng et al., 2025). HousingQA questions are statutory-entailment questions framed in landlord-tenant terms, sharing vocabulary with the housing statutes that answer them, so raw queries are already corpus-shaped. The main paper reports both benchmarks (Zheng et al., 2025).

BarExamQA. BarExamQA contains 1,195 multiple-choice bar-exam questions in the evaluation split used here. The retrieval corpus is an unstructured collection of legal passages, with some passages labeled ‘gold passages’ by legal experts that are intended to be the ideal evidence for answering one of the questions (Zheng et al., 2025). Hit@5 and MRR@5 metrics therefore measure whether the retriever exposes the labeled rule passage before the answer model chooses an option.

HousingQA. HousingQA contains 6,853 yes/no statutory entailment questions over state housing statutes (Zheng et al., 2025). The corpus contains many jurisdictions, and each question names the relevant state. All main HousingQA retrieval rows apply that state as a metadata filter before ranking.

4.2. Models, Baselines, and Retrieval Stack

We evaluate three model configurations: Llama 3.1 8B Instant, Gemma 4 26B, and Llama 3.3 70B Versatile. Within each response the same model serves as both the query generator (when one is needed) and the final answer model, so accuracy lift cannot be attributed to a stronger reader. The route labels in the released records map to Llama 3.1 8B Instant via Groq (groq-llama8b), Gemma 4 26B via OpenRouter (or-gemma4-26b, pinned to the logged same-model provider route), and Llama 3.3 70B Versatile via Groq (groq-llama70b).

We compare four deployable retrieval methods and providing models with the gold passage. Models using retrieval methods answer from $k = 5$ retrieved passages.

- LLM only answers the original question without retrieval.
- Raw question RAG retrieves with the question text, then answers from the question plus top- k evidence.
- HyDE retrieves with a generated legal passage, without a draft answer prior.
- Snap-HyRE (ours) retrieves with a draft-guided legal search passage; the draft answer is private and is not shown to the final answer call.
- Gold Evidence supplies the single labeled gold passage directly.

Structured rewrite and gold-plus-neighbor (using gold passage to retrieve 4 more passages) diagnostics appear in the appendix; route labels are retained in the released artifacts for reproducibility.

Retrieval stack. Runs use seed 42, temperature 0, and a 2048-token answer cap. If the model produces a malformed generated passage or no parseable final answer, the example is retried with the same model/provider route; final-answer repairs are format-only when a parseable prediction already exists, and rows fail closed under `NO\_SILENT\_FALLBACK`. The scored answer is the last exact final line in the required format. Retrieval uses Chroma collections built with Alibaba-NLP/gte-large-en-v1.5 embeddings; a cross-encoder (ms-marco-MiniLM-L-6-v2) then reranks the candidate set before the top- k cut.

Generated-query evaluation. Generated-query methods are evaluated in two stages. First, the query-generation call writes the HyDE passage or Snap-HyRE search passage. Then retrieval stores the ranked document ids. The answer run replays those ids, and that evidence is used in final question answering.

Per-dataset retrieval interfaces. BarExamQA retrieval uses the shared fact pattern and question stem; generated methods may see option text but no correctness marker. HousingQA retrieval applies a state metadata filter. The filter is not a Snap-HyRE-specific feature; it is applied to every HousingQA row to prevent retrieval of evidence from a different jurisdiction.

4.3. Answer and Evidence Metrics

The downstream metric is exact answer accuracy. BarExamQA uses the last valid multiple-choice answer line, and HousingQA uses Answer: Yes or Answer: No. Retrieval is measured by Hit@5 and MRR@5: whether any labeled support passage appears in the top five, and how early it appears.

We also report efficiency as the number of model calls per question, the tokens the answer call processes per question (the “answer-stage tokens”), and correct answers per million answer-stage tokens. Two-call methods are not penalized in this count: the first-stage query-generation call is counted as part of the method footprint and logged separately. The answer-stage token count shows how much context the final answer call sees once retrieval has selected evidence.

The main tables include only validated full complete conditions. Appendix A.1 lists the inclusion rule, coverage grid, supplemental controls, top- k diagnostics, efficiency table, and reproducibility details.

5. Results

5.1. When Snap-HyRE Helps Answers

Snap-HyRE helps in proportion to the question-corpus vocabulary gap. On BarExamQA, fact patterns describe accidents and transactions in narrative prose while the support corpus uses doctrinal language; raw-question RAG retrieves the labeled rule passage in only 1.4% of cases at Hit@5. Snap-HyRE raises Hit@5 to 9.5%, 12.1%, and 11.0% across the 8B, 26B, and 70B models, and converts those retrieval gains into answer gains of +2.4pp, +4.0pp, and +5.1pp over raw-question RAG; the Gemma 4 26B improvement is paired-significant at McNemar $p < 0.001$. The answer gain grows with model size, while retrieval exposure is not strictly monotone: the Gemma 4 26B row has the highest Hit@5.

HousingQA sits at the opposite end of that axis. Statutory-entailment questions already share vocabulary with the target housing statutes; raw-question retrieval reaches 36.9% Hit@5 with no query generation. With less of a gap to close, Snap-HyRE trades a small answer margin for retrieval exposure: it trails raw-question RAG on the Llama rows and reaches 65.1% on Gemma 4 26B, essentially tied with HyDE at 65.0% but below raw-question RAG at 66.1%. The same Gemma 4 26B Snap-HyRE row improves Hit@5 from 36.9% to 38.1%. Section 5.6 reports an $N = 500$ Gemma 26B prompt diagnostic that improves retrieval with statistically unchanged paired answer accuracy on that slice.

5.2. Did the Right Evidence Appear?

Table 2 separates retrieval from answer selection. Figure 3 extends that view across k from one to ten for both datasets. On BarExamQA, generated legal passages move every top- k metric up: Snap-HyRE and HyDE both raise Hit@5 by roughly an order of magnitude over raw retrieval, with Snap-HyRE slightly ahead at every depth. On HousingQA, raw queries remain strong because they already share vocabulary with the corpus. The generated-query average sits below the state-filtered raw retrieval baseline, but the Gemma 26B Snap-HyRE row is the strongest generated-query retrieval row and exceeds raw at Hit@5 and MRR@5.

Snap-HyRE converts an average +9.5pp Hit@5 and +4.8pp MRR@5 retrieval shift into a +3.9pp average BarExamQA answer accuracy gain over raw-question RAG, while exact gold evidence still appears in only about one in nine top-five lists. The remaining gap points to ranking, option mapping, and evidence use.

Table 1. Main answer accuracy (%). Bold marks the largest value in each column among the deployable methods. The Gold Evidence row—where we provide the model the gold passage—is set apart by a rule and is not eligible for bolding. Averages are descriptive over available cells.

Method	BarExamQA				HousingQA			
	8B	26B	70B	Avg.	8B	26B	70B	Avg.
LLM	57.3	80.8	78.7	72.3	55.4	56.1	44.8	52.1
Raw question RAG	54.5	78.0	74.6	69.0	62.3	66.1	62.1	63.5
HyDE	56.1	80.3	80.2	72.2	59.1	65.0	62.2	62.1
Snap-HyRE (ours)	56.9	82.0	79.7	72.9	59.0	65.1	59.6	61.2
Gold Evidence	60.0	78.6	79.2	72.6	64.3	—	67.3	65.8

Table 2. Gemma 26B evidence exposure at $k = 5$ (%). Bold marks the largest value in each column. HousingQA uses jurisdiction state filtering for all retrieval methods; this table reports retrieval exposure, not answer accuracy.

Method	BarExamQA		HousingQA	
	Hit@5	MRR@5	Hit@5	MRR@5
Raw question RAG	1.4	0.7	36.9	23.3
HyDE	11.4	5.4	30.6	19.6
Snap-HyRE (ours)	12.1	6.0	38.1	24.5

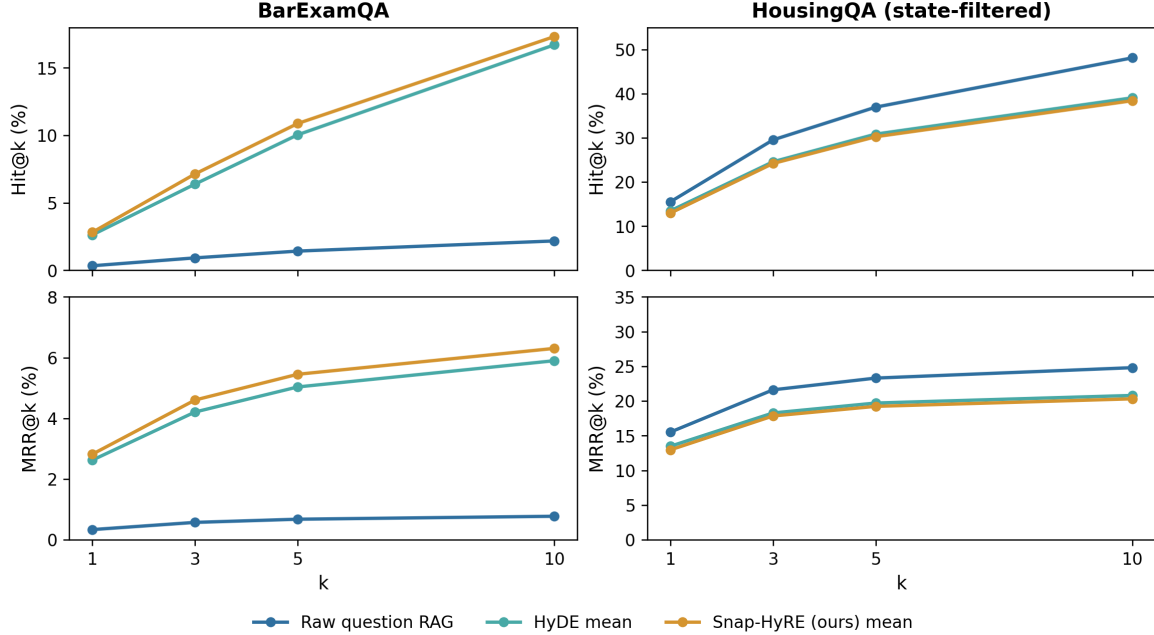


Figure 3. Top- k retrieval exposure and rank quality for full top-10 caches. BarExamQA and HousingQA generated-query curves average over Llama 3.1 8B, Gemma 4 26B, and Llama 3.3 70B; HousingQA uses the jurisdiction state filter.

5.3. Differentiation from Previous Methods

On this item, raw-question RAG retrieves generic accident material and HyDE drifts toward proximate cause without landing on the rule. Snap-HyRE writes a short passage about foreseeable intervening injury, surfaces the MBE proximate-cause rule, and picks (B). The win comes from a more directed search query than HyDE or the raw question produces.

5.4. Snap-HyRE Matches the Gold Evidence on Larger Models

On BarExamQA, Snap-HyRE exceeds the single labeled-gold-passages control on Gemma 4 26B (82.0% versus 78.6%) and approximately matches it on Llama 3.3 70B (79.7% versus 79.2%). Only Llama 3.1 8B trails the gold evidence method. The retrieved top-5 list does more work than the labeled rule alone: surrounding legal passages supply context the answer

Table 3. Worked BarExamQA example (mbe_1175, Gemma 4 26B). Gold answer: B (driver liable for both injuries).

Method	Pred.	Top-5 retrieval
Raw question RAG	D ✗	Generic accident and intervening-cause material; gold rule passage missed.
HyDE	D ✗	Drifts toward proximate cause but stays broad; gold rule passage missed.
Snap-HyRE (ours)	B ✓	Anchored on foreseeable subsequent injury; surfaces MBE proximate-cause rule passages.

Table 4. Answer-stage token accounting over five logged cells per method. First-stage query-generation usage is excluded. Bold marks the best single-call non-gold evidence retrieval method.

Method	In/q	Out/q	Correct/M
LLM	131	237	1487.2
Raw RAG	2295	339	241.5
HyDE	2237	340	244.5
Snap-HyRE	2062	338	258.8
Gold	1447	215	404.5

model uses. The same pattern shows up at the gold evidence level in Appendix A.2: adding neighbors to the gold passage also improves accuracy on Llama 3.1 8B and Gemma 4 26B.

5.5. Leading Answer-Token Efficiency

Table 4 reports correct answers per million tokens that the answer call processes. Among non-gold evidence methods Snap-HyRE is the most answer-stage context-efficient at 258.8 correct per million answer-stage tokens, versus 244.5 for HyDE and 241.5 for raw-question RAG: the reranked short passages produce a shorter average answer context at higher accuracy. The first-stage generation tokens are excluded from this answer-stage accounting. The gold evidence method reaches 404.5 correct per million tokens because it sees only the single labeled passage.

5.6. One-Shot Corpus Inclusion

The first call’s prompt can be made more corpus-shaped with a single example. We add one real passage from the target corpus to the Snap-HyRE generation prompt (with its question, answer, and document id stripped) and re-run the first call. On Gemma 4 26B $N = 500$ probes, +Exemplar improves canonical Snap-HyRE retrieval on both main datasets: BarExamQA moves from 13.0% to 13.6% Hit@5, and state-

Table 5. Statute/passage exemplar lift for Gemma 4 26B $N = 500$ retrieval probes (Hit@5 / MRR@5, %). Δ is the exemplar lift over canonical Snap-HyRE. Bold marks the higher of Snap-HyRE vs. +Exemplar within each metric per row.

Dataset	Snap-HyRE	+ Exemplar	Δ Hit@5 / MRR@5
BarExamQA	13.0 / 6.4	13.6 / 6.3	+0.6 / -0.1
HousingQA	38.2 / 24.3	41.2 / 26.5	+3.0 / +2.2

filtered HousingQA moves from 38.2% to 41.2% Hit@5 with MRR@5 rising from 24.3% to 26.5%. On HousingQA, +Exemplar also beats raw-question retrieval on Hit@5. The paired HousingQA answer slice is statistically unchanged (62.8% with the example versus 63.0% for plain Snap-HyRE, $p = 1.0$), so the result is best read as a prompt-level retrieval improvement rather than an answer-accuracy claim. On an Ohio writ-of-execution question, for example, the prompted passage names “writ of restitution” and “execution” in statute style and retrieves the gold Ohio statute at rank 3, where plain Snap-HyRE drifts to neighboring writ-of-possession language.

6. Conclusion

Snap-HyRE encodes a small piece of expert legal reasoning into a fixed two-call pipeline: form an intuition about the controlling doctrine, retrieve evidence by that frame, then judge the evidence on its own without anchoring to the intuition. The lift over raw-question RAG tracks the question-corpus vocabulary gap, because that gap is precisely the regime in which the intuition step matters. When the raw question is a poor retrieval handle, letting the model form an intuition before it searches and forcing the final answer back to the corpus delivers both retrieval and answer gains; when the question already reads like a passage from the corpus, HousingQA shows raw-question RAG remains the stronger answer baseline, while the Gemma 26B full-corpus retrieval row and one-example diagnostic show where targeted prompt conditioning can still recover evidence exposure.

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A. Appendix

A.1. Result Inclusion Policy

A row is used as a paper claim only when the result record names the dataset, provider, method, answer depth, metric values, and any row note needed to interpret the result. Rows without validated full results are omitted from the main comparison rather than imputed.

- Main benchmarks: BarExamQA and HousingQA.
- Route labels: groq-llama8b, or-gemma4-26b, groq-llama70b.
- Main modes: llm_only, rag_simple, rag_hyde, snap_hyre, and golden_passage. rag_rewrite is a supplemental generated-query control, and golden_plus_neighbors is retained only as a neighbor-dilution diagnostic.
- Answer depth: $k = 5$ for retrieval rows.
- HousingQA interface: retrieval rows must use the jurisdiction state filter to count as main-interface rows.

The main two-benchmark matrix reports 35 validated cells out of 42 expected (BarExamQA 21/21; HousingQA 14/21); missing cells are left blank rather than imputed.

Table 6. Coverage of the BarExamQA and HousingQA main matrix in this snapshot (validated rows over expected rows for the seven canonical methods across three models).

Dataset	Validated	Expected
BarExamQA	21	21
HousingQA	14	21
Total	35	42

Table 7. Tagged row notes. These tags mark validation conditions to carry with row-level claims; they do not by themselves remove a row from the included set or imply severity. Counts are non-exclusive because a row may carry more than one note.

Tagged rows	Note
31	Format-validation note
17	Output-length note
5	Standard row
3	Endpoint-identity note
2	Auxiliary-metadata note
2	Segmented-evaluation note

A.2. Supplemental Controls

The supplemental tables separate the fixed-method comparison from controls that are useful for interpreting failure modes: structured rewrites, gold evidence context, and neighbor dilution.

Table 8. Dataset-level and pooled deltas. Mean percentage-point differences across signed model slices.

Slice	Snap vs. raw pp	HyDE vs. Snap pp	Gold vs. raw pp
BarExamQA	+3.8 (n=3)	-0.7 (n=3)	+3.6 (n=3)
HousingQA	-2.3 (n=3)	+0.9 (n=3)	+3.6 (n=2)
Pooled	+0.8 (n=6)	+0.1 (n=6)	+3.6 (n=5)

Table 10. Gold evidence controls. Accuracy values are percentages; deltas are percentage points over included method pairs. Bold marks the highest accuracy in each row across Raw question RAG, Snap-HyRE, and Gold.

Dataset	Model	Raw question RAG	Snap-HyRE	Gold	Gold vs. raw pp
BarExamQA	Llama 3.1 8B	54.5	56.9	60.0	+5.5
BarExamQA	Gemma 4 26B	78.0	82.0	78.6	+0.6
BarExamQA	Llama 3.3 70B	74.6	79.7	79.2	+4.6
HousingQA	Llama 3.1 8B	62.3	59.0	64.3	+2.0
HousingQA	Llama 3.3 70B	62.1	59.6	67.3	+5.2

Table 9. Snap-HyRE against non-gold evidence controls. Cells are accuracy percentages; bold marks the strongest performing model-method pair.

Dataset	Model	LLM	Raw question RAG	HyDE	Snap-HyRE	Rewrite	Snap vs. raw pp	Best main non-gold evidence
BarExamQA	Llama 3.1 8B	57.3	54.5	56.1	56.9	57.3	+2.4	LLM / Rewrite (57.3)
BarExamQA	Gemma 4 26B	80.8	78.0	80.3	82.0	80.7	+4.0	Snap-HyRE (ours) (82.0)
BarExamQA	Llama 3.3 70B	78.7	74.6	80.2	79.7	77.2	+5.1	HyDE (80.2)
HousingQA	Llama 3.1 8B	55.4	62.3	59.1	59.0	—	-3.3	Raw question RAG (62.3)
HousingQA	Gemma 4 26B	56.1	66.1	65.0	65.1	—	-1.1	Raw question RAG (66.1)
HousingQA	Llama 3.3 70B	44.8	62.1	62.2	59.6	—	-2.5	HyDE (62.2)

Table 11. Neighbor-dilution diagnostic for the gold evidence method. Positive deltas mean retrieved passages (using gold passage as query) helped beyond the providing just the gold passage to the model. Bold marks the higher of Gold vs. Gold + neighbors per row.

Dataset	Model	Gold	Gold + neighbors	Δ pp
BarExamQA	Llama 3.1 8B	60.0	61.8	+1.8
BarExamQA	Gemma 4 26B	78.6	80.7	+2.1
BarExamQA	Llama 3.3 70B	79.2	77.8	-1.4

A.3. Top-k Retrieval Diagnostics

Tables 12 and 13 give the underlying per-row retrieval exposure as the cutoff grows from one to ten passages; the same data drives Figure 3 in Section 5.

Table 12. HousingQA raw retrieval with and without the jurisdiction filter. Values are percentages from complete top-10 caches. Bold marks the higher value in each column.

Corpus scope	Hit@1	Hit@3	Hit@5	Hit@10	MRR@10
National corpus	0.8	1.9	2.8	5.1	1.8
Jurisdiction metadata filter	15.5	29.5	36.9	48.1	24.8

BarExamQA generated passages raise the chance of seeing the labeled evidence throughout the top-10 list, although MRR remains modest because the exact supporting passage is rarely ranked first. HousingQA behaves differently: the jurisdiction filter is the major retrieval correction, and raw state-filtered questions already retrieve relevant evidence often.

Table 13. Top- k retrieval diagnostics from complete top-10 caches. Values are percentages; mean rows average model-specific generated-query caches. Bold marks the highest value in each metric column within each (dataset $\times$ model/scope) slice.

Dataset	Model/scope	Method	n	Hit@3	Hit@5	Hit@10	MRR@10
BarExamQA	Shared	Raw question RAG	1195	0.9	1.4	2.2	0.8
BarExamQA	Mean over 3 models	HyDE	1195	6.4	10.0	16.7	5.9
BarExamQA	Mean over 3 models	Snap-HyRE (ours)	1195	7.1	10.9	17.3	6.3
BarExamQA	Llama 3.1 8B	HyDE	1195	5.5	8.3	13.5	5.2
BarExamQA	Llama 3.1 8B	Snap-HyRE (ours)	1195	6.3	9.5	14.8	5.4
BarExamQA	Gemma 4 26B	HyDE	1195	7.1	11.4	19.1	6.4
BarExamQA	Gemma 4 26B	Snap-HyRE (ours)	1195	7.9	12.1	18.7	6.9
BarExamQA	Llama 3.3 70B	HyDE	1195	6.5	10.5	17.6	6.1
BarExamQA	Llama 3.3 70B	Snap-HyRE (ours)	1195	7.2	11.0	18.5	6.6
HousingQA	State-filtered shared	Raw question RAG	6853	29.5	36.9	48.1	24.8
HousingQA	Mean over 3 full models	HyDE	6853	24.6	30.8	39.1	20.8
HousingQA	Mean over 3 full models	Snap-HyRE (ours)	6853	24.2	30.2	38.4	20.3
HousingQA	State-filtered Llama 3.1 8B	HyDE	6853	21.6	27.0	34.5	17.9
HousingQA	State-filtered Llama 3.1 8B	Snap-HyRE (ours)	6853	23.4	29.6	37.0	19.9
HousingQA	State-filtered Gemma 4 26B	HyDE	6853	24.3	30.6	40.1	20.9
HousingQA	State-filtered Gemma 4 26B	Snap-HyRE (ours)	6853	31.1	38.1	47.0	25.7
HousingQA	State-filtered Llama 3.3 70B	HyDE	6853	27.9	34.9	42.6	23.6
HousingQA	State-filtered Llama 3.3 70B	Snap-HyRE (ours)	6853	18.1	23.1	31.3	15.3

A.4. Statute/Passage Exemplar Probe Details

The exemplar variant adds one fixed sanitized real-passage exemplar to the Snap-HyRE generation prompt, with the exemplar’s own question, answer, and document id stripped. The same $N = 500$ retrieval interfaces are used as the corresponding canonical Snap-HyRE rows: BarExamQA uses the shared support-passage collection; HousingQA uses the jurisdiction state filter. The retrieval probes are computed directly from the cached passage-id lists; paired answer accuracy was scored only on the HousingQA $N = 500$ slice (62.8% exemplar vs 63.0% canonical Snap-HyRE, $-0.20pp$, $p = 1.0$). Table 5 in Section 5.6 reports the lift values.

A.5. Prompt and Route Reproducibility

All generated-query rows use deterministic decoding (temperature 0) and the same model for query generation and final answering. The route labels are groq-llama8b for Groq Llama 3.1 8B Instant, or-gemma4-26b for OpenRouter Gemma 4 26B (google/gemma-4-26b-a4b-it, with same-model provider route logged), and groq-llama70b for Groq Llama 3.3 70B Versatile. Snap-HyRE first calls require two parseable blocks, a draft answer and a retrieval passage; HyDE first calls require only the retrieval passage. Final answer calls score the last exact Answer: line. Malformed generation or answer format is retried only under the same model/provider route, and silent fallback is disabled.

A.6. Reproducibility Statement

All tables and figures are derived from the result records used for this draft. The source package includes paper-facing tables and figure data; the full experiment logs, retrieval caches, and plotting scripts are retained in the project repository paths named in those records and should be released with a manifest for exact regeneration.